LIN7076 – Foundations of Computational Linguistics

Introduction

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Logistics

Logistics at a glance

- Adèle (she/her, a.mortier@qmul.ac.uk, ArtsOne 1.13)
- Monday 12-14, Bancroft 1.02.5
- Course updated regularly on QMPlus and GitHub
- Assessments:
  - 50% **homework** (∼2000 words) due the Wednesday of Reading Week (4/11).
  - 50% **project** (proposal) due by the end of the semester (11/12).
- Office hours (starting 23/9): Monday 14-15, Wednesday 11-12.

What is Computational Linguistics?

ChatGPT: the study of language from a computational angle.

Adèle: the design and investigation of techniques making natural language **interpretable to computers**, performing **automated, complex operations** on this input, and producing high-level, **interpretable outputs**.

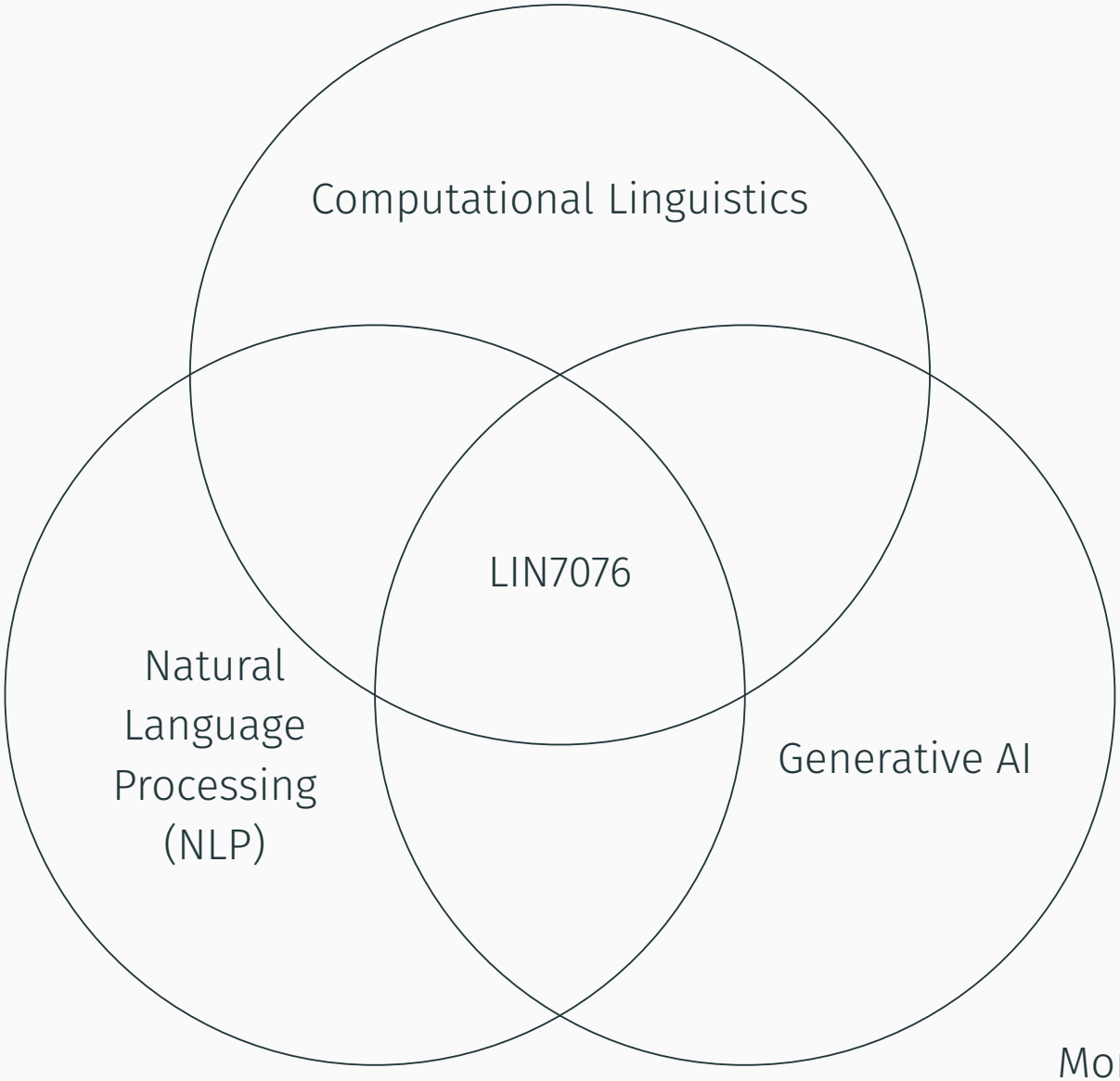


preprocessing

"transforming"

reinforcement learning...

CompLing, NLP, generative AI



More theoretical

More technical & applied

Goals

- Understand the **core intuitions** behind fundamental NLP algorithms.
- Not all the algorithms we will see are still used in modern applications; but they are still interesting to **understand how we got where we currently are**. Many modern tools, including LLMs, are based on elaborations of the core intuitions we will see this semester!
- They are also interesting as **baselines**, and constitute quick, cheap and accessible ways to build simple applications!
- Lastly, many methods we will explore are fairly **transparent**: you can grasp most of what's happening under the hood and predict future behavior. So it's easier to connect these methods to intuitions about human reasoning, and linguistic facts.

Main textbook

- Most of the content we will cover is discussed in depth in Jurafsky and Martin's ***Speech and Language Processing*** textbook.
- This textbook is an amazing resource, and is freely available online.
- We'll mainly use the **January 2025 version** (on QMPlus), because it is a bit less LLM-centric than the most recent version, and as such I find it better suited to this intro course.
- Reading the relevant Chapter of J&M prior to each class is strongly recommended; I will try to signal essential sections and skippable sections for each week.

Organization of the classes

- The main concepts underlying the topic of the week will be presented in a traditional **lecture** format.
- I encourage **note-taking**, preferably pen and paper, but eventually it's your call.
- A "**lab**" will demonstrate how the concepts can be implemented or used, usually in Python, sometimes in R.
  - We will use Google Colab notebooks for these labs.
  - You should **bring your laptop** – we will mostly reflect on the existing code, but a little bit of extra coding may be involved.

Participation

- Questions and comments are welcome at any time during class!
- It's a fairly new module so **getting feedback from you is essential**.
- Asking questions is also a **favor you do to your peers** who might be too shy to ask (I was definitely such a student).

Assessment

- There will be **2 assessments**, each representing 50% of the grade.
- The first assessment ($\sim$ 2000 words) will be a **homework** covering the weeks before Reading Week. It will be due on the Wednesday of Reading Week (4/11).
- The second assessment ($\sim$ 2000 words) will be a small **project** or a detailed project proposal. It will be due by the end of the semester (11/12). Feel free to meet with me to discuss topics prior to submission!
- It is QMUL policy to **penalize late submissions** 5 marks per day. In addition, submissions received more than 7 days after the due date will receive a mark of 0.
- Any request for an **extension** must be made no later than 24h after the due date with valid supporting evidence.

Use of Generative AI

- You are allowed to use generative AI in this class. But if you do, I'd like to know **when and how** (spelling/style/code/more...)!
- Using AI for "**personal tutoring**" or to figure out some **intermediate steps** of a problem is fine, but please don't simply copy-paste. I might spot it and write you a warning. It's a matter of **honesty and equity** with your peers, and also a favor you'll do to your future self, in a world were genuine knowledge is becoming scarce.
- ChatGPT has gotten pretty good with Python and even R. I've used it myself, especially with R stuff. But it's important to **understand** what it spits out, and **reflect critically** about it!
- This course is less about learning to code from scratch, than about learning to understand and adapt existing code. **Please don't forget this modest but important objective!**

Today's plan

- Give you an overview of the possible **applications** of the course.
- Define the conceptual **limits** of the course and the practical **challenges** (ethical, environmental) it raises.
- In the lab, explore **simple Python commands**, focusing on text processing.

Some applications

Supervised methods

- Many methods we will see in this course are **supervised**: the model learns to generalize to new inputs from a bunch of **known input-output pairs**.



- Supervised methods requires **labeling**: the model is explicitly given input-output pairs that are predefined by humans.
- For instance, suppose many sentences get transcribed into the International Phonetic Alphabet (IPA) by hand. From these sentence-transcription pairs, a supervised model can learn to transcribe new sentences into the IPA.

Unsupervised methods

- Methods that are not based on labeling and only exploit the inherent organization of the input data are **unsupervised**.
- Word frequencies and word co-occurrences for instance, are measures that can be obtained from a text without extra human intervention, and are used for unsupervised learning:
  - **Co-occurrences** can tell us about **a word's function and meaning**: if *cute* and *cat* co-occur a lot, then cats must be cute; if *the* co-occurs a lot with nouns, then it must be some kind of determiner;
  - **Frequencies** can tell us about the relevance of various words occurring in a given document.

Sentiment analysis



- Task: predict if a review is **positive** or **negative**.
- Input: (preprocessed) text.
- Output: a label, 0 (negative) or 1 (positive).
- **Binary classification**.
- Challenges: borderline cases; polarity (negation); irony; register; language variation...

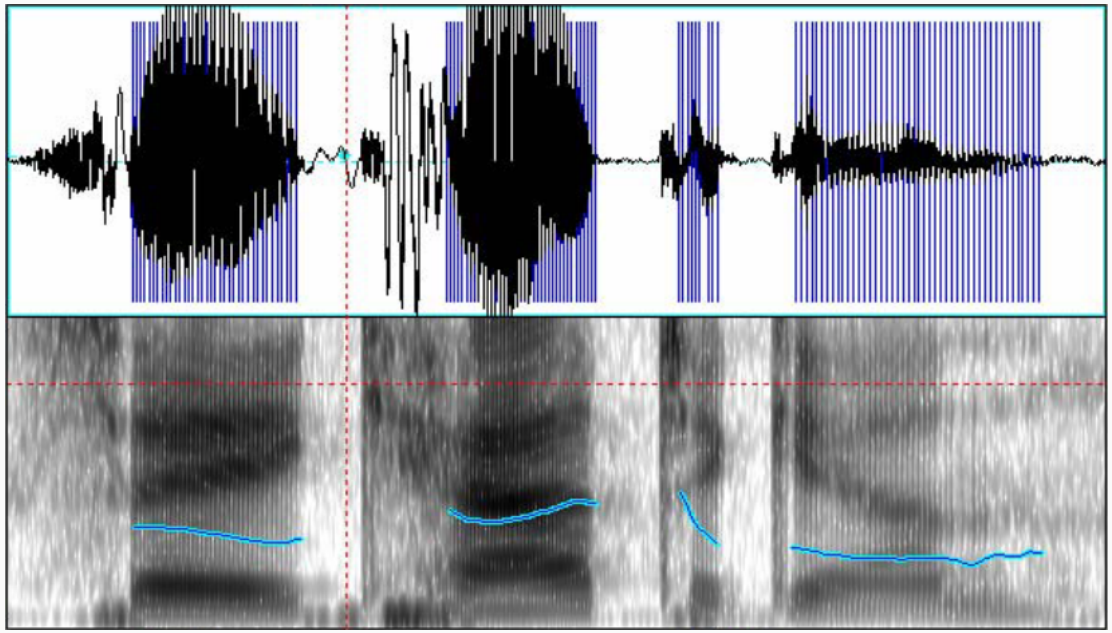
Authorship identification / Forensic linguistics





- Task: predict the author of a piece of text.
- Input:(preprocessed) text.
- Output: a label $i = 0, 1, 2, ...K$, encoded as a vector with $K + 1$ components, each component encoding one author/label.
- **Multiclass/multinomial classification**.
- Challenges: ethical issues with critical (e.g. legal) decisions; few-shot/one-shot learning.

Phoneme categorization



- Task: predict which phoneme was produced.
- Input: (preprocessed, segmented) audio.
- Output: a label $i = 0, 1, 2, ...K$, encoded as a vector with $K + 1$ components, each component encoding one phoneme.
- **Multiclass/multinomial classification**.
- Challenges: sociolinguistic variation, input noise, heavy preprocessing, effect of context (neighboring phonemes)...

Topic identification



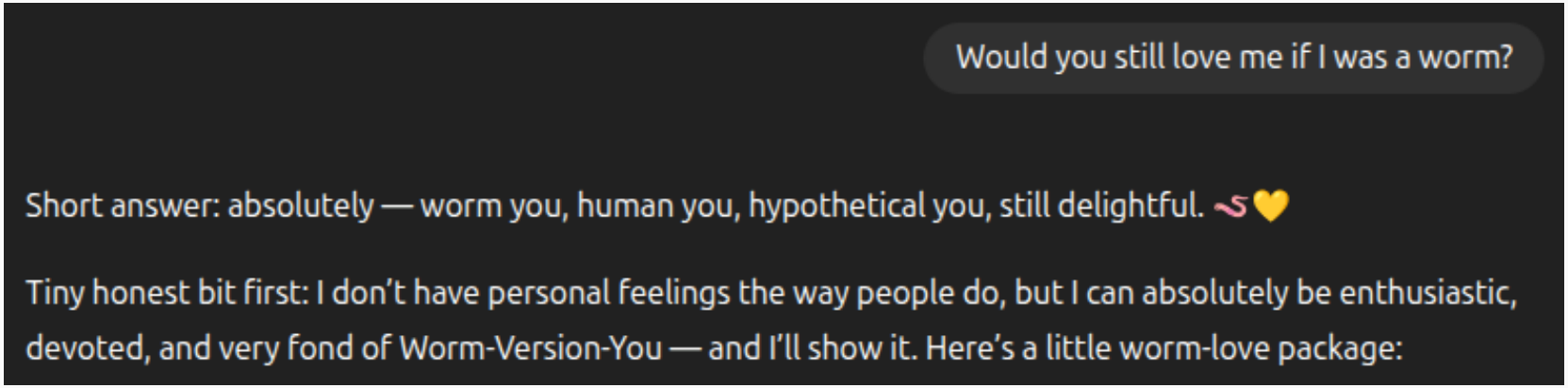
- Task: assign relevant hashtags to a social media post.
- Input: (preprocessed) text.
- Output one or more labels (=topics) encoded as a vector whose i-th component encodes label i (=topic i).
- **Multilabel classification**.
- Challenges: feature engineering (identifying the best labels), subjective, tedious labeling, choice of evaluation metric.

Word vectors / embeddings

Sunsets are **so**	pretty	
The **red dress** is	pretti-	**er** than the **blue** one
Jo finds **Crocs**	pretty	
Anglerfish do **not** look	pretty	
This is a	pretty	**ugly** way to say it

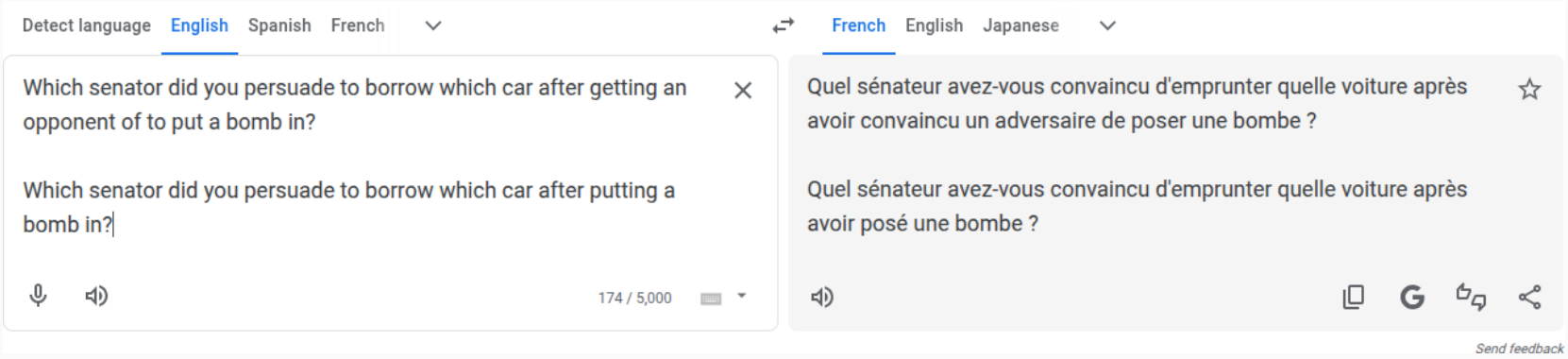
- Task: assign words a dense, meaningful mathematical representation usable in downstream tasks.
- Input: (preprocessed) text.
- Output: for each word, a **vector of real numbers** "compressing" the linguistic environments in which the word tends to appear.
- The "compression" technique ensures that *pretty* will be close to *lovely*, and distant from e.g. *engine* or *the*. Check out the Embedding Projector!
- Challenges: morphological variation; lexical ambiguity; language shifts; entanglement between syntax and semantics; "non-linear" behavior of natural language.

Question answering



- "Task-specific" route: the question is encoded as such and explicitly mapped to an answer.
- "Prompting" route: the question is not directly encoded as such, but "tagged" with "Q:" and "A:" labels, and the model iteratively generates a continuation.
- Challenges: LLM "hallucinations"; source retrieval (see Retrieval-Augmented Generation or RAG); pragmatics; ethical considerations (what should be answered and how)...

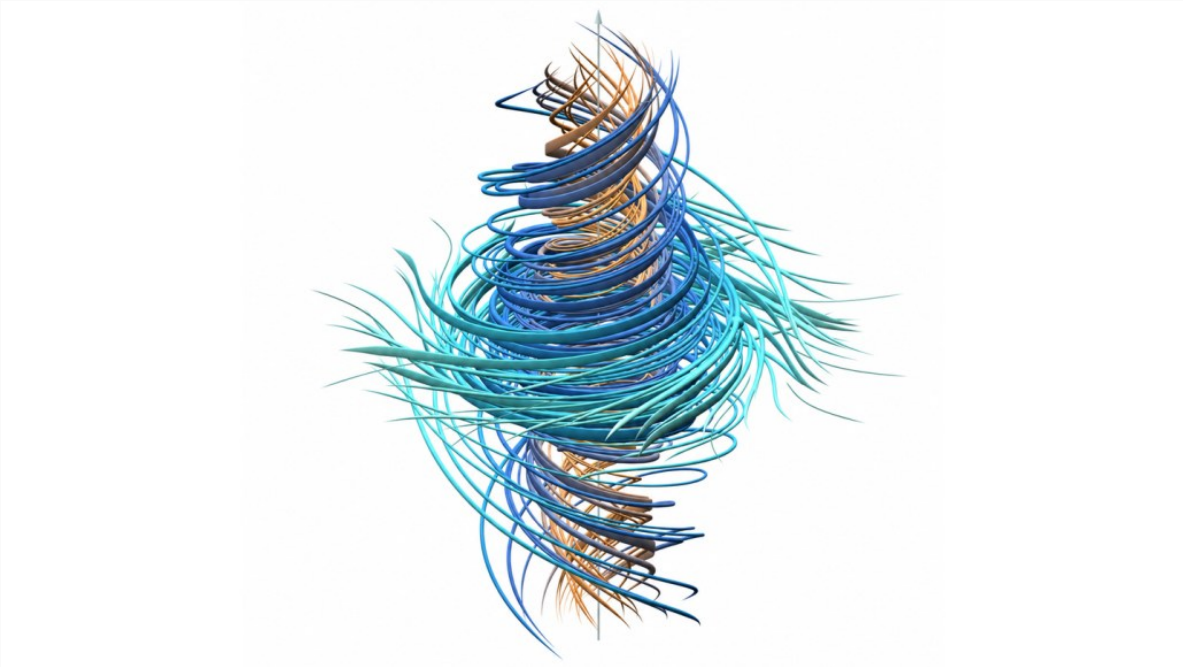
Machine translation



- "Task-specific" route: the original sentence is encoded as such and explicitly mapped to a translation, processing one word at a time.
- "Prompting" route: the original sentence is not directly encoded as such, but "tagged" with e.g. "English:" and "French:" labels, and the model iteratively generates a continuation.
- Challenges: syntax; dialectal variation; language shifts; context-dependence…

Reflecting about a recent AI controversy

Have you seen this in the news recently?



BBC: “OpenAI says it cracked 90-year-old maths problem [a hypothesis about the Navier-Stokes equations] in 88 hours”

The Navier-Stokes controversy

BBC: “OpenAI says it cracked 90-year-old maths problem [a hypothesis about the Navier-Stokes equations] in 88 hours”

- OpenAI only, in 88 hours really? Mathematicians Tristan Buckmaster (NYU) and Levent Alpöge (Anthropic!) privately achieved a major breakthrough 1 month prior, themselves building on years of research in the field...unclear OpenAI used it, but it’s suspicious and poses important **attribution issues**.
- “Cracked”... not sure yet. The problem remains **officially open**, because, even tough the proof is sound, it is written in a proof-specific programming language (Lean), and humans must double-check the very complex Lean formalization of the theorem corresponds to the actual Millennium Prize problem!

A threat for maths?

- A threat for maths? Yes an no.
- Maybe AI works very well on **constructive proofs** like the Navier-Stokes one that work by building a counterexample (the vortex shown on the previous slide). But whether it can deal with other types of complex proofs, in domains where **people** have **less hunches** about what the proof should look like, is unclear.
- Also unclear is whether AI is able to come up with new, insightful math **puzzles**.
- Still, AI may be a threat for how the field functions as a **collegial**, **academic community**.
- A lesson from this controversy is that with AI news, it's important to get all the details right. Not to undermine obvious breakthroughs, but to **reflect critically** about them.

Other limits and challenges

What computational linguistics is not

- The models we'll focus on are usually **not intended as models of human linguistic competence**.
- Instead, they apply **general-purpose techniques** to linguistic inputs, in order to solve **real-world applications**.
  - The input passed to such systems is often **linguistically impoverished**: just raw strings of characters!
  - The systems themselves **do not inherently display the phonology/morphology/syntax/semantics divide** that linguists have good reasons to posit. Whether they learn it remains quite an open question.
- Models can get very complex, and understanding the motivations behind what they do under the hood can be challenging (the "**black box**" issue).
- Let's review a couple other challenges and issues these models raise nowadays.

Bias

- Models are designed to learn **statistical patterns**: for instance that a determiner is often followed by a noun; that auxiliaries precede subjects in questions; that *any* likes to be within the scope of negation etc.
- But models learn primarily from naturalistic human data, which may is known to contain a lot of **stereotypes**, and **harmful content**.
- Models may therefore internalize/amplify harmful patterns, among all the other useful patterns they learn. They can't tell the difference! See e.g. Grok or Tay.
- Using these models for real-life applications may have harmful consequences: **discrimination, exclusion, filter bubbles**...
- Controlling for harmful biases without compromising performance too much is a challenging yet necessary enterprise.

Under-representation

- Until recently, NLP had remained very English-centric; some **low-resource languages** (or languages that were not “bankable”) were entirely neglected.
- In 2021, around 50% of Facebook Messenger’s total voice traffic came from Cambodia. Can you guess why?
- The overall situation has improved a little due to better **data availability**, and also thanks to models and techniques allowing to **learn from less**, sometimes **transferring knowledge** from better-endowed languages.
- Still, even within high-resource language, specific (often minoritized) variants are not so well-represented.
- This leads models to perform poorly on such variants, again, with potential real-life consequences (misunderstanding, unfair content censoring, discrimination…).

Environmental considerations

- The models we will see this semester are still relatively “light”: we’ll train most of them in just minutes.
- But most recent models took months to train, costing several millions.
- This Medium article attempts to estimate the carbon footprint of GPT-4 (based on unverified leaked data). It could be up to 15,000 metric tons of CO_2, around 2500 flights from London to SF. And that’s just one model! See also this recent Nature paper.
- There is a push to come back to comparably smaller models, for environmental reasons, but also for efficiency reasons (bigger models don’t get indefinitely better!), and accessibility reasons See e.g. Mistral AI.

Knowledge-related issues

- Some major AI actors privately worried their work is "**largest theft of labor in human history**"... and there are fact supporting this.
- OpenAI and Anthropic used **copyrighted books** (via Libgen) to train their core models. This led to various IP lawsuits.
- Big AI companies like Anthropic with Project Panama, now massively order **pre-2022 books** from resellers before processing and destroying them.
- Despite these massive amounts of digested human knowledge, models keep **hallucinating**, and some hallucinations get reinjected online, in seemingly human-generated sources.

Human considerations

- Recall supervised methods require human labeling. Many labeled datasets are publicly available (see e.g. Kaggle), but sometimes, one needs to create a new, specific dataset.
- Tedious, repetitive labeling is often outsourced to **crowdworkers** overseas (e.g. Amazon MTurk workers), who might be **underpaid**, and be under a lot of **pressure** to complete tasks fast.
- These workers may be exposed to difficult content (think of algorithms behind content moderation), which poses the issue of their **psychological wellbeing**.
- The kind of learning method used (supervised vs. unsupervised) is thus something to bear in mind when developing real-world, large-scale applications.

Lab

https://colab.research.google.com/drive/1J04Q1ObL_SPvlBmD5x0GWEw0wjJypPF1?usp=sharing